

NEET Important Chapter – Systems of Particles and Rotational Motion

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In the previous chapter, we understood the motion of a single particle which has no size and point mass. In this chapter, we will encounter a body of finite size and a system of particles and try to understand the motion of a system as a whole. In this case, the centre of mass plays an important role. Thereafter, we will discuss the motion of the centre of mass (position, velocity, acceleration, etc.) and learn the concept of rigid body, rotational motion, kinematics and dynamics of rotational motion about a fixed axis and rolling motion.

We will also learn about moment of force (torque), angular momentum of a particle, angular acceleration, torque and angular momentum for a system of particles, conservation of angular momentum, angular velocity, linear momentum of the system of particles and its relationship with linear velocity in this chapter.

At the end, we will discuss the equilibrium of a rigid body and centre of gravity. We will also deal with an important concept of moment of inertia and theorems of perpendicular and parallel axes. We will compare the translational motion and rotational motions and also discuss how to apply Newton's equation in rotational motion.

We will also define the moment of inertia of various objects such as a ring, disc, cylinder, solid sphere, etc. In this article, we will cover the concepts which are useful for the NEET exam and boost your exam preparation.

Important Topics of Systems of Particles and Rotational Motion

- Rigid Bodies
- Rotational Kinematics
- Rolling motion
- Parallel Axis Theorem
- Perpendicular Axis Theorem
- Torque
- Angular Momentum
- Angular Displacement
- Conservation of Angular Momentum

Important Concepts of Systems of Particles and Rotational Motion

Name of the Concept	Key Points of the Concepts
1. Rigid body	The point at which the whole mass of bodies is supposed to be concentrated in order to study the motion of a body according to Newton's law of motion is known as the centre of mass of a body. Rigid bodies undergo mainly three types of motion: Pure Rotational, Pure Translational and both Translational or Rotational.
2. Angular motion	When a body rotates about a fixed axis, the rotation is called a rotatory motion or angular motion and the axis is called the axis of rotation
3. Torque	It is defined as the algebraic sum of moments of the force acting on all the particles of the rotating rigid body about the axis of rotation. SI unit of torque is Newton -metre

4. Angular momentum	It is defined as the sum of moments of all the particles of a rotating rigid body about the axis of rotation.
5. Moment of inertia	Moment of inertia of the body about a given axis is the sum of products of the masses of the particles and squares of their distances from the axis of rotation. It is defined as: Where, r_i - The perpendicular distance of particle of mass m_i from the axis of rotation Unit - — Moment of inertia in rotational motion is analogous to mass of the body in translational motion. Moment of inertia is also known as rotational inertia.
5. Law of conservation of angular momentum	It states that the angular momentum of a rotating body about an axis remains constant if no external torque is applied to it. I.e.,
6. Radius of gyration	Radius of gyration of the body about a given axis of rotation is the distance of the axis of rotation and the point at which the whole of the mass of the body is supposed to be concentrated so as to have the same moment of inertia.
7. Perpendicular axes theorem	The moment of inertia about a uniform plane about an axis perpendicular to its plane is equal to the sum of its moment of inertia about any two perpendicular axes in the plane that intersect at a point, where the perpendicular axis cuts the lamina.

8. Parallel axis theorem	Moment of inertia of a body about any axis is equal to the moment of inertia about the parallel axis passing through the centre of mass plus product of the mass of the body and square of the perpendicular distance between these two parallel axes. Where, d- The perpendicular distance between the two axes.
9. Rotational equilibrium	A body is said to be in rotational equilibrium if the resulting torque acting on it is zero.

List of Important Formulae

Sl. No.	Name of Concept	Formula
1.	Moment of Inertia of ring	
2.	Moment of inertia of disc	_____
3.	Moment of inertia of a solid sphere	—
4.	Moment of inertia of spherical shell	—
5.	Moment of inertia of hollow cylinder	
6.	Moment of inertia of thin rod of length L	_____
7.	Moment of inertia of a Solid cylinder:	_____
8.	Torque	
9.	Radius of gyration	_____ _____
10.	Rotational work and energy	
11.	Rotational kinetic energy	—
12.	Angular momentum	
13.	Angular impulse	

Comparison of Rotational Motion and Linear Motion and Some important Relations

- Moment of inertia in rotational motion is analogous to mass of the body in translational motion.
- Angular displacement (θ) in rotational motion is analogous to linear displacement of the body in translational motion.

Linear Displacement:

Angular Displacement:

- Angular velocity in rotational motion is analogous to linear velocity of the body in translational motion.

Linear velocity: $v = r\omega$

Angular velocity: $\omega = \frac{v}{r}$

Relation between linear velocity and angular velocity:

- Angular acceleration in rotational motion is analogous to linear displacement of the body in translational motion.

Linear acceleration: $a = r\alpha$

Angular acceleration: $\alpha = \frac{a}{r}$

- Torque in rotational motion is analogous to linear force of the body in translational motion.

Relation between torque and force:

- Equations of motion -

S.No.	Translational motion	Rotational motion
1. First equation of motion		
2. Second equation of motion	—	—
3. Third equation of motion		

Solved Examples of System of Particles and Rotational Motion

1. Find the torque of a force about the origin. The force acts on a particle whose position vector is .

Sol:

Given:

Position vector =

And force =

The relation between torque and force:

By determinant rule,

Key Point – Remember the relation which relates the torque and force and apply it with care of direction.

2. Two rings have their moment of inertia in the ratio 4:1 and their radii are in the ratio 4:1. Find the ratio of their masses .

Sol:

Given, the ratio of moment of inertia of two rings is 4:1

The ratio of the diameters of two rings is 4:1

We know that the moment of inertia of the ring is $I =$

Key Point: Apply the formula of moment of inertia of the ring and find the ratio of masses.

Previous Year Questions of System of Particles and Rotational Motion

1. Find the torque about the origin when force of _____ acts on a particle whose position vector is _____. (NEET 2020)

Sol:

Given:

Position vector =

Force =

The relation between torque and force:

Trick: Apply the formula of torque that is relation between torque and force and take care of directions of position vector and force. Cross product of _____ and _____ is - _____ which is used in the above question.

2. From a circular ring of mass 'M' and radius 'R', an arc corresponding to a 90 degree sector is removed. The moment of inertia of the remaining part of the ring about an axis passing through the centre of the ring and perpendicular to the plane of the ring is 'K' times . Then the value of 'K' is: (NEET 2021)

Sol:

Given, mass of the ring = M

Radius of the ring = R

We know that moment of inertia of the circular ring is $I = MR^2$

According to the question, an arc of 90 degree is removed from the ring that is one fourth part of the original ring, so that mass is also one fourth of the mass of the original ring (M).

So, the moment of inertia of the removed part is $\frac{1}{4}MR^2$

Now, the moment of inertia of the remaining part of the circular ring is $\frac{3}{4}$ of the moment of inertia of ring that is $\frac{3}{4}MR^2$

The moment of inertia of the remaining part of the ring about an axis passing through the centre of the ring and perpendicular to the plane of the ring is 'K' times $\frac{3}{4}MR^2$

By comparison -

Value of k is $\frac{3}{4}$.

Trick: Ring is a symmetric shape and density is uniform throughout the ring, so mass is also distributed uniformly.

Practice Questions

1. A horizontal circular plate is rotating about a vertical axis passing through its centre with an angular velocity ω . A man sitting at the centre having two blocks in his hands stretches out his hands so that the moment of inertia of the system doubles. If the kinetic energy of the system is K initially, its final energy will be (**Ans: $\frac{K}{2}$**)

2. Two particles A and B initially at rest, move towards each other under mutual force of attraction. At the instant when the speed of A is V and the speed of B is $2V$, the speed of the centre of mass of the system is (**Ans: Zero**)

3. A particle undergoes uniform circular motion. About which point on the plane of the circle will the angular momentum of the particle remain conserved ? (**Ans: Centre of the circle**)

Conclusion

In this article, we have discussed one of the important chapters for NEET, System of Particles and Rotational Motion. We have covered most of the important concepts, formulae and solved examples along with previous year questions of the NEET exams. In this article, we have also studied all the parameters and characteristics required for understanding rotational kinematics and rotary motion. This article will help students to prepare for the [NEET 2022](#) exam and revise all the topics together.

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